# The Burning Platform

Our credit portfolio earns a 2.1% true economic return and a single product category destroys $450K annually

**2.1%**
True Economic Return
(after charge-off losses)

**$28.2M**
Permanently Destroyed
(Unrecovered Losses)

**13.82%**
Bad Loan Rate
(Portfolio Average)

## Portfolio Economics

| | |
|---|---|
| Total Applications | 38,576 |
| Total Funded | $435.7M |
| Total Received | $473.1M |
| Gross Net Return | +$37.3M |
| Unrecovered Loss | –$28.2M |
| True Econ Return | ~$9.1M |
| True Return Rate | 2.1% |
| Bad Loans | 5,333 |
| Recovery Rate | 56.90% |

*Source: financial_loan.csv | 38,576 records | $435.7M portfolio*

METHODOLOGY



# Analytical Approach & Data Foundation

38,576 records · $435.7M funded · 24 variables · 9 analytical dimensions · financial_loan.csv

### SQL Server

Data ingestion, 9 analytical KPI queries, derived metric creation (LTI Ratio, Concentration Score), cross-validation layer for all DAX measures

### Power BI

3-page interactive dashboard: Executive Command Centre | Risk Diagnostic Engine | Profitability Lens. 7 custom DAX measures.

### Excel

Cross-validation of all SQL outputs, pivot table verification, scenario stress-testing, final presentation data tables

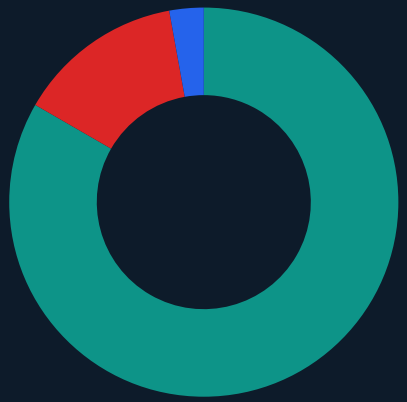
## The 9 Analytical Lenses

1. Grade Risk Pricing
2. DTI Escalation Curve
3. Verification Paradox
4. Small Business Crisis
5. Purpose × Term Matrix
6. Employment Length Myth
7. LTI Cliff (Derived)
8. Geographic Concentration
9. Grade × Purpose Map

# $435.7M Funded · $37.3M Earned · $28.2M Permanently Destroyed

| 38,576 | $435.7M | $473.1M | +$37.3M | 13.82% | $28.2M |
|---|---|---|---|---|---|
| Total Applications | Total Funded | Total Received | Gross Net Return | Bad Loan Rate | Unrecovered Loss |

## Loan Status Distribution



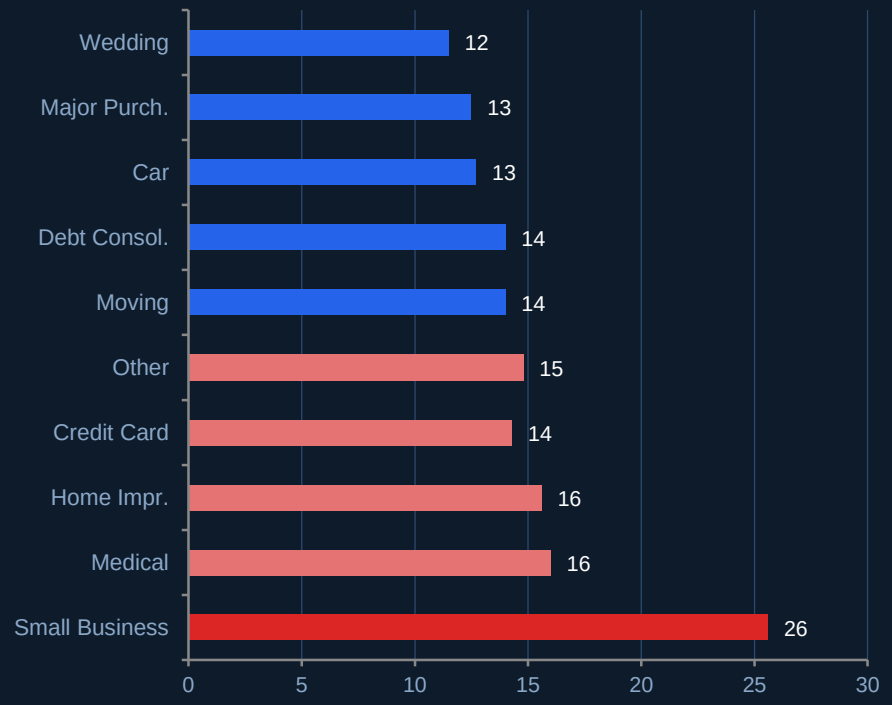
Fully Paid (83.3%) Charged Off (13.8%) Current (2.8%)

## Default Loss Analysis

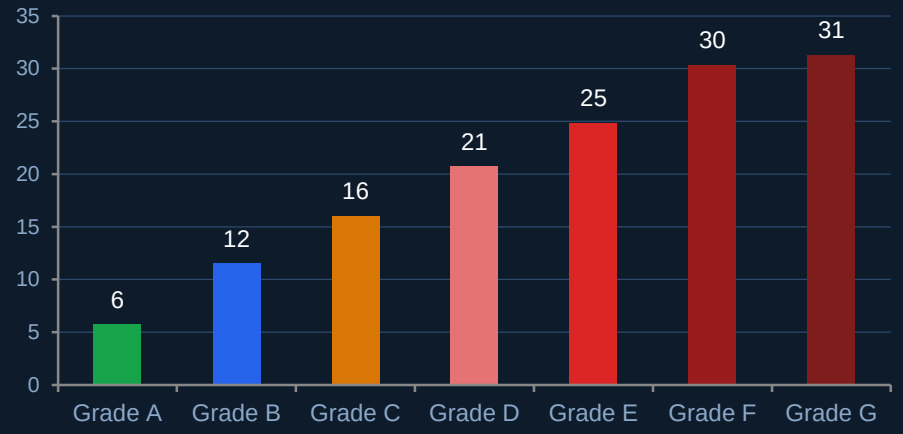
| Metric | Value |
|---|---|
| Total Applications | 38,576 |
| Capital at Risk (Bad Loans) | $65,532,225 |
| Recovered from Defaults | $37,284,763 |
| Unrecovered True Loss | $28,247,462 |
| Recovery Rate on Defaults | 56.90% |
| True Economic Return Rate | ~2.1% |

# Default Risk is Not Random — It Concentrates in Specific Segments

### Default Rate % by Loan Purpose




### Default Rate % by Grade




KEY FINDING

**Small Biz: 25.6% default — 92% worse than portfolio average**

Grade G reaches 31.3% — nearly 1 in 3 loans defaults
Portfolio average: 13.82%

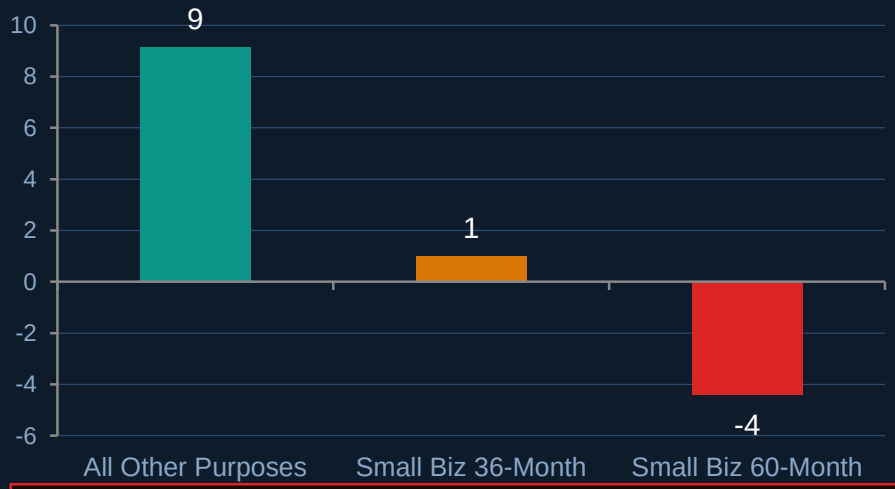
# Small Business 60-Month Loans: A Confirmed Loss Product

**34.55%**
Default Rate
(60M Small Biz)

**–$449K**
Net Loss on
$10.2M Funded

**25.62%**
Overall Small Biz
Default Rate

## Net Return % by Segment




## Full Breakdown

| Segment | Default % | Net Ret % |
|---|---|---|
| All Others (36,800) | 13.26% | +9.14% |
| All Small Biz (1,776) | 25.62% | –1.28% |
| Small Biz 36M (1,203) | 21.36% | +1.01% |
| **Small Biz 60M (573)** | **34.55%** | **–4.42%** |

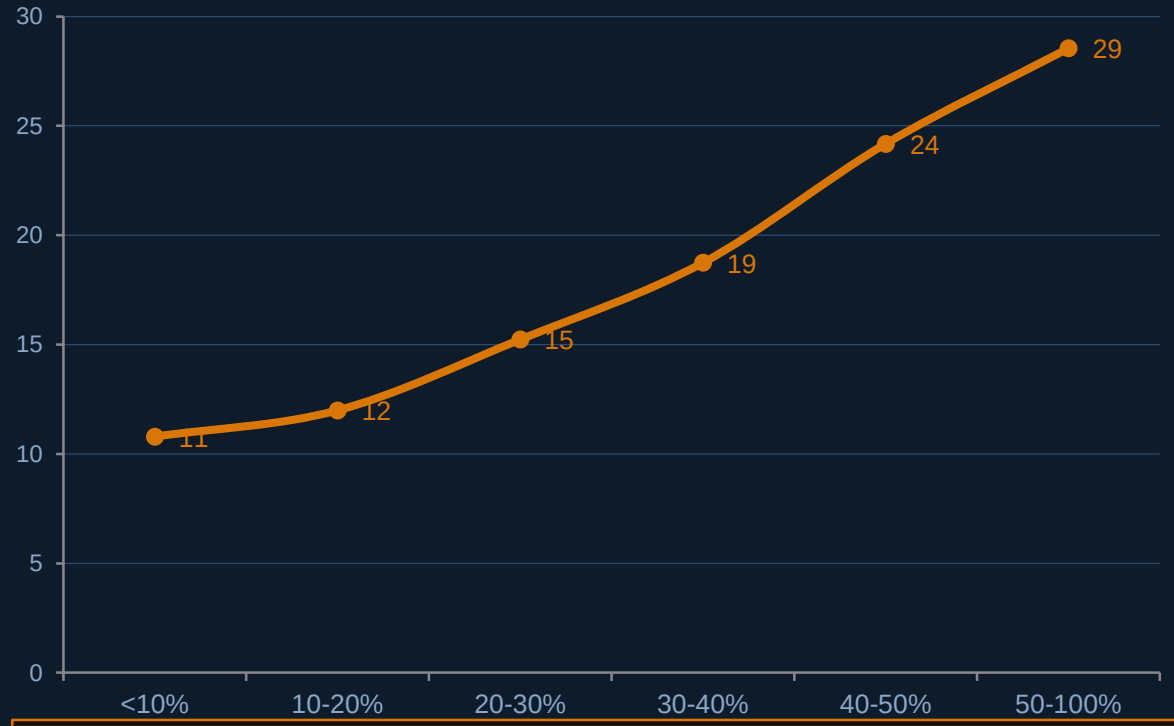
**RECOMMENDATION: Immediately suspend 60-month small business origination → estimated +$450K annual recovery**

NOVEL DERIVED INSIGHT



# The Loan-to-Income Cliff: A New Risk Dimension

Borrowers taking a loan worth >30% of annual income default at 18.7% — 74% worse than sub-10% borrowers




## How to Read This

**LTI < 10%**
10.79% default. Safe zone.

**LTI 10–20%**
11.99% — minimal escalation.

**LTI 20–30%**
15.23% — crosses portfolio avg.

**LTI 30–40%**
18.74% — THE CLIFF. +74% vs baseline.

**LTI 40–50%**
24.18% — 1 in 4 defaults. Severe.

**LTI 50%+**
28.54% — nearly 1 in 3 defaults.

**RECOMMENDATION: Implement LTI hard cap at 35% in underwriting criteria → estimated +$2–4M annual risk reduction**

MYTH-BUSTING INSIGHT



# The Verification Paradox

Verified borrowers default 3.46 pp MORE — because verification tracks loan size, not creditworthiness

## NOT VERIFIED

**12.24%**

Default Rate

| | |
|---|---|
| Avg Loan | **$8,485** |
| Avg Income | **$61,607** |
| Avg Int Rate | **11.23%** |

## SOURCE VERIFIED

**14.14%**

Default Rate

| | |
|---|---|
| Avg Loan | **$10,136** |
| Avg Income | **$68,811** |
| Avg Int Rate | **12.12%** |

## VERIFIED

**15.70%**

Default Rate

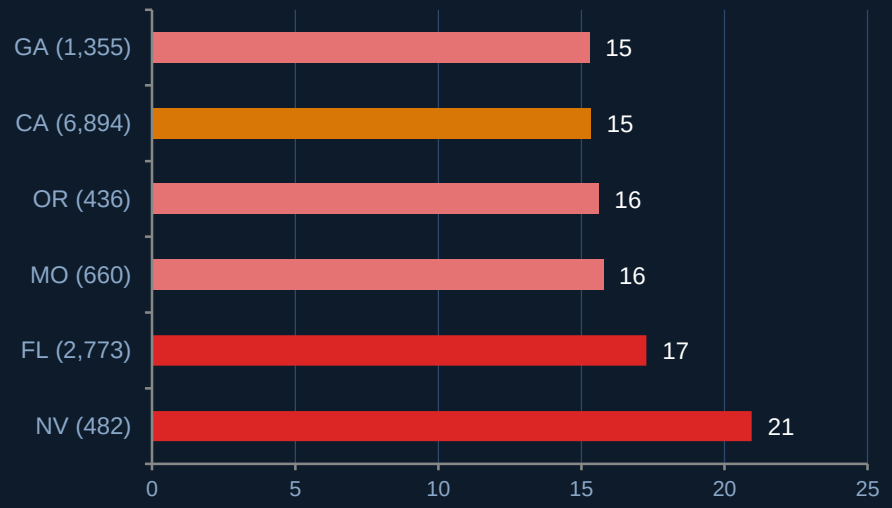
| | |
|---|---|
| Avg Loan | **$15,968** |
| Avg Income | **$81,034** |
| Avg Int Rate | **13.09%** |

EXPLANATION: Verification is triggered by loan SIZE not risk. Verified = $7,483 more borrowed on avg. Variable measures exposure, not safety.

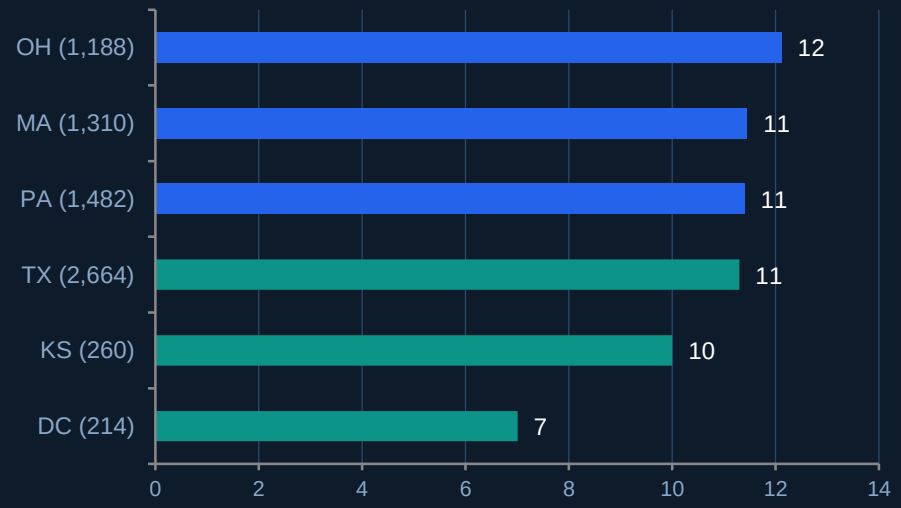
# California: 18% of Portfolio at 15.33% Default — Single-State Concentration Risk

| 13.94 pp | 18% | 20.95% | 7.01% |
|---|---|---|---|
| Gap: Best vs Worst State | California Portfolio Share | Nevada Default Rate (Highest) | DC Default Rate (Lowest) |

**Highest Default States (min. 200 loans)**




**Lowest Default States (min. 200 loans)**



# Grade × Purpose: Where the Bank is Paying to Lend

5 grade-purpose combinations with negative net returns · $396K destroyed on $16M funded

| | Debt Consol. | Credit Card | Small Biz | Home Imp. | Medical | Other |
|---|---|---|---|---|---|---|
| Grade A | +5.2% | +5.8% | +7.2% | +6.1% | +4.5% | +5.0% |
| Grade B | +7.8% | +9.2% | -3.9% | +7.5% | +5.8% | +6.3% |
| Grade C | +9.5% | +11.0% | +6.2% | +8.8% | +7.2% | +8.1% |
| Grade D | +10.8% | +12.4% | -2.2% | +9.5% | +8.3% | +9.2% |
| Grade E | +11.2% | +13.0% | -0.8% | +10.1% | +9.0% | +9.8% |
| Grade F | +11.1% | +12.8% | -1.1% | +10.5% | +9.5% | +10.2% |

NET RETURN %: NEGATIVE (Loss) · < 5% · 5–9% · > 9% Target

# Three Policy Changes · Estimated $5–8M Annual Value Recovery

## 01 Suspend 60-Month Small Business Loans

IMMEDIATE

Product generates –4.42% net return with 34.55% default rate. Only confirmed loss product in the portfolio.

↑ ~+$450K annual recovery

LOW — credit policy change only, no technology.

## 02 Implement LTI Cap at 35%

30 DAYS

Borrowers above 35% LTI default at 18.74%+. Derived metric not in raw data — requires new origination system field.

↑ ~+$2–4M risk reduction

MEDIUM — requires LTI field in underwriting workflow.

## 03 Redesign Small Business Scorecard

60-90 DAYS

Small biz fails across ALL grade levels. Needs purpose-specific model: min. 2 yrs in business, collateral, co-signer.

↑ ~+$1–2M annual recovery

HIGH — new credit model and underwriting guidelines.

# 30–60–90 Day Implementation Plan

| | Days 1–30 \| Quick Wins | Days 31–60 \| Policy Build | Days 61–90 \| Model Redesign |
|---|---|---|---|
| **Risk Team** | Suspend 60M small business origination immediately | Validate LTI vs credit bureau data for accuracy | Back-test small business scorecard on historical portfolio |
| **Credit Policy** | Issue policy memo: 60M small business moratorium | Draft LTI cap policy — 35% hard limit in guidelines | Publish new small business underwriting handbook |
| **Tech / Systems** | Flag 60M SB pipeline; block new applications in system | Build LTI calculation field in origination workflow | Deploy new scorecard model in origination system |

# Technical Appendix — Supporting Analytics

## Data Profile

38,576 rows | 24 columns | $435.7M funded | financial_loan.csv | Jan–Feb 2021 window | Zero nulls in key fields | Decimal DTI (0.20 = 20%)

## SQL Validation

Master portfolio query + 9 insight queries | All KPIs cross-validated against Power BI DAX | Reproducible in MSSQL Server, PostgreSQL, or any ANSI SQL

## Derived Metrics

LTI Ratio = loan_amount / annual_income | Concentration Risk Score = portfolio_share% × default_rate% | True Economic Return = Gross Return − Unrecovered Loss

## Analytical Assumptions

Bad Loan = Charged Off only | Net Return = total_payment − loan_amount | Minimum 50 loans for segment credibility | DTI stored as decimal in source data

## Power BI DAX Measures

Bad Loan Rate % | Net Return | Net Return % | LTI Ratio | Unrecovered Loss | Recovery Rate % | True Economic Return | 3-page dashboard

## What to Investigate Next

FICO score at origination | Macro conditions at vintage | Post-charge-off collections by state | Repeat borrower behaviour | Income verification accuracy